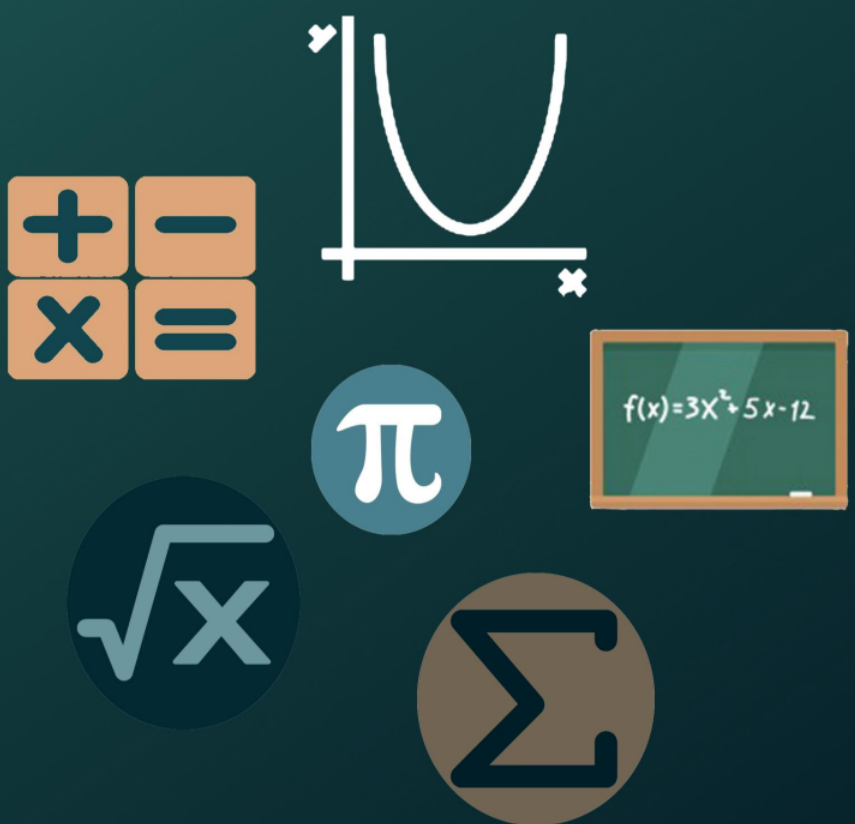
MATHEMATICS

# Matrix




SHEET

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***Note***

*Detailed solutions are available on the eSaral App.*

# *Module Description*




For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

*"Practice makes a man perfect"*

**1. Concept builder – 1 & 1A**

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

**2. Brain Booster – 2 & 2A**

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

**3. Simulator – JM & JA**

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

# *Home Work Index*

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

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## Exercise - 1 — Concept Builder

### SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

### Types of Matrices

**Q1.** If $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 \\ 4 & 5 & 6 & 0 \\ 7 & 8 & 9 & 10 \end{bmatrix}$, then A is

(A) An upper triangular matrix
(B) A null matrix
(C) A lower triangular matrix
(D) None of these

**Q2.** Square matrix $[a_{ij}]_{n\times m}$ will be an upper triangular matrix, if

(A) $a_{ij} \neq 0$, for $i > j$ (B) $a_{ij} = 0$, for $i > j$

(C) $a_{ij} = 0$, for $i < j$ (D) None of these

**Q3.** In a upper triangular matrix $n \times n$, minimum number of zeros is

(A) $n(n-1)/2$
(B) $n(n+1)/2$
(C) $2n(n-1)/2$
(D) None of these

**Q4.** In order that the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 3 & \lambda & 5 \end{bmatrix}$ be non-singular, $\lambda$ should not be equal to

(A) 1 (B) 2
(C) 3 (D) 4

**Q5.** If $\begin{bmatrix} 2+x & 3 & 4 \\ 1 & -1 & 2 \\ x & 1 & -5 \end{bmatrix}$ is a singular matrix, then $x$ is

(A) $\frac{13}{25}$ (B) $-\frac{25}{13}$

(C) $\frac{5}{13}$ (D) $\frac{25}{13}$

### Algebra of Matrices

**Q6.** If $2A + 2B = \begin{bmatrix} 2 & -1 & 4 \\ 3 & 2 & 5 \end{bmatrix}$ and $A + 2B = \begin{bmatrix} 5 & 0 & 3 \\ 1 & 6 & 2 \end{bmatrix}$, then $2B =$

(A) $\begin{bmatrix} 8 & -1 & 2 \\ -1 & 10 & -1 \end{bmatrix}$ (B) $\begin{bmatrix} 8 & 1 & 2 \\ -1 & 10 & -1 \end{bmatrix}$

(C) $\begin{bmatrix} 8 & 1 & -2 \\ -1 & 10 & -1 \end{bmatrix}$ (D) $\begin{bmatrix} 8 & 1 & 2 \\ 1 & 10 & 1 \end{bmatrix}$

**Q7.** $A = \begin{bmatrix} 5 & -3 \\ 2 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 6 & -4 \\ 3 & 6 \end{bmatrix}$, then $A - B =$

(A) $\begin{bmatrix} 11 & -7 \\ 5 & 10 \end{bmatrix}$ (B) $\begin{bmatrix} -1 & 1 \\ -1 & -2 \end{bmatrix}$

(C) $\begin{bmatrix} 11 & 7 \\ 5 & -10 \end{bmatrix}$ (D) $\begin{bmatrix} 12 & -7 \\ 5 & -10 \end{bmatrix}$

**Q8.** If $\begin{bmatrix} x & 0 \\ 1 & y \end{bmatrix} + \begin{bmatrix} -2 & 1 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 6 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 4 \\ 2 & 1 \end{bmatrix}$, then

(A) $x = -3, y = -2$ (B) $x = 3, y = -2$
(C) $x = 3, y = 2$ (D) $x = -3, y = 2$

**Q9.** If $2X - \begin{bmatrix} 1 & 2 \\ 7 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 0 & -2 \end{bmatrix}$, then X is equal to

(A) $\begin{bmatrix} 2 & 2 \\ 7 & 4 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 2 \\ 7/2 & 2 \end{bmatrix}$

(C) $\begin{bmatrix} 2 & 2 \\ 7/2 & 1 \end{bmatrix}$ (D) None of these

**Q10.** If $\begin{bmatrix} x+y & 2x+z \\ x-y & 2z+w \end{bmatrix} = \begin{bmatrix} 4 & 7 \\ 0 & 10 \end{bmatrix}$, then values of x, y, z , w are

(A) 2, 2, 3, 4 (B) 2, 3, 1, 2

(C) 3, 3, 0, 1 (D) None of these

## Multiplication of Matrices

**Q11.** $\cos\theta \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} + \sin\theta \begin{bmatrix} \sin\theta & -\cos\theta \\ \cos\theta & \sin\theta \end{bmatrix} =$

(A) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

(C) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

**Q12.** If $R(t) = \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}$, then $R(s).R(t) =$

(A) $R(s)+R(t)$ (B) $R(st)$

(C) $R(s+t)$ (D) None of these

**Q13.** If $A = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$

and $B = \begin{bmatrix} \cos\beta & -\sin\beta \\ \sin\beta & \cos\beta \end{bmatrix}$,

then the correct relation is

(A) $A^2 = B^2$ (B) $A + B = B - A$

(C) $AB = BA$ (D) None of these

**Q14.** If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ a & b & -1 \end{bmatrix}$, then $A^2 =$

(A) Unit Matrix

(B) Null Matrix

(C) A

(D) –A

**Q15.** Let $A = \begin{bmatrix} 5 & 5\alpha & \alpha \\ 0 & \alpha & 5\alpha \\ 0 & 0 & 5 \end{bmatrix}$. If $|A^2| = 25$, then $|\alpha|$ equals

(A) $5^2$ (B) 1

(C) 1/5 (D) 5

**Q16.** If $A = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix}$ and $A^n = O$, then the minimum value of n is

(A) 2 (B) 3

(C) 4 (D) 5

**Q17.** If $A = \begin{bmatrix} 1/3 & 2 \\ 0 & 2x-3 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 6 \\ 0 & -1 \end{bmatrix}$ and $AB = I$, then $x =$

(A) –1 (B) 1

(C) 0 (D) 2

**Q18.** If AB = C, then order of matrices A, B, C can be-

(A) $A_{2\times3}, B_{3\times2}, C_{2\times3}$

(B) $A_{3\times2}, B_{2\times3}, C_{3\times2}$

(C) $A_{3\times3}, B_{2\times3}, C_{3\times3}$

(D) $A_{3\times2}, B_{2\times3}, C_{3\times3}$

**Q19.** If $A = [1\ 2\ 3]$ and $B = \begin{bmatrix} -5 & 4 & 0 \\ 0 & 2 & -1 \\ 1 & -3 & 2 \end{bmatrix}$, then

$AB =$

(A) $\begin{bmatrix} -5 & 4 & 0 \\ 0 & 4 & -2 \\ 3 & -9 & 6 \end{bmatrix}$ (B) $\begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$

(C) $\begin{bmatrix} -2 & -1 & 4 \end{bmatrix}$ (D) $\begin{bmatrix} -5 & 8 & 0 \\ 0 & 4 & -3 \\ 1 & -6 & 6 \end{bmatrix}$

**Q20.** If $A = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} -5 & 7 & 1 \\ 1 & -5 & 7 \\ 7 & 1 & -5 \end{pmatrix}$

then AB is equal to

(A) $I^3$ (B) $2I_3$

(C) $4I_3$ (D) $18I_3$

**Q21.** If $A = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 0 & 3 \\ -2 & 2 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \\ 5 & -4 & 0 \end{bmatrix}$,

then the element of third row and third column in AB will be

(A) –18

(B) 4

(C) –12

(D) None of these

**Q22.** If $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, then $A^5 =$

(A) 5A (B) 10A

(C) 16A (D) 32A

**Q23.** If $A$ and $B$ are two matrices and $(A+B)(A-B) = A^2 - B^2$, then

(A) $AB = BA$

(B) $A^2 + B^2 = A^2 - B^2$

(C) $A'B' = AB$

(D) None of these

**Q24.** If $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, then $A^4 =$

(A) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

(C) $\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

**Q25.** If $A = dig(2,-1,3), B = dig(-1,3,2)$, then

$A^2B =$

(A) dig (5, 4, 11) (B) dig (–4, 3, 18)

(C) dig (3, 1, 8) (D) B

**Q26.** If $P = \begin{pmatrix} i & 0 & -i \\ 0 & -i & i \\ -i & i & 0 \end{pmatrix}$ and $Q = \begin{pmatrix} -i & i \\ 0 & 0 \\ i & -i \end{pmatrix}$,

then $PQ$ is equal to **(where i =** $\sqrt{-1}$**)**

(A) $\begin{pmatrix} -2 & 2 \\ 1 & -1 \\ 1 & -1 \end{pmatrix}$ (B) $\begin{pmatrix} 2 & -2 \\ -1 & 1 \\ -1 & 1 \end{pmatrix}$

(C) $\begin{pmatrix} 2 & -2 \\ -1 & 1 \end{pmatrix}$ (D) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

**Q27.** If $M = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$ and $M^2 - \lambda M - I_2 = 0$, then

$\lambda =$

(A) –2 (B) 2

(C) –4 (D) 4

**Q28.** Let three matrices are $A=\begin{bmatrix}2 & 1\\ 4 & 1\end{bmatrix}$;

$B=\begin{bmatrix}3 & 4\\ 2 & 3\end{bmatrix}$ and $C=\begin{bmatrix}3 & -4\\ -2 & 3\end{bmatrix}$,

then $t_r(A) + t_r\left(\frac{ABC}{2}\right) + t_r\left(\frac{A(BC)^2}{4}\right) +$

$t_r\left(\frac{A(BC)^3}{8}\right) + \ldots\ldots + \infty$ is equal to-

(A) 6 (B) 9

(C) 12 (D) none

**Q29.** For a matrix $A=\begin{bmatrix}1 & 2r-1\\ 0 & 1\end{bmatrix}$, the value of

$\prod_{r=1}^{50}\begin{bmatrix}1 & 2r-1\\ 0 & 1\end{bmatrix}$ is equal to -

(A) $\begin{bmatrix}1 & 100\\ 0 & 1\end{bmatrix}$ (B) $\begin{bmatrix}1 & 4950\\ 0 & 1\end{bmatrix}$

(C) $\begin{bmatrix}1 & 5050\\ 0 & 1\end{bmatrix}$ (D) $\begin{bmatrix}1 & 2500\\ 0 & 1\end{bmatrix}$

## Transpose of a Matrix

**Q30.** $A=\begin{bmatrix}4 & 6 & -1\\ 3 & 0 & 2\\ 1 & -2 & 5\end{bmatrix}$, $B=\begin{bmatrix}2 & 4\\ 0 & 1\\ -1 & 2\end{bmatrix}$, $C=\begin{bmatrix}3\\ 1\\ 2\end{bmatrix}$,

then the expression which is not defined is

(A) $A^2 + 2B - 2A$ (B) $CC'$

(C) $B'C$ (D) $AB$

**Q31.** The product of a matrix and its transpose is an identity matrix. The value of determinant of this matrix is :

(A) – 1 only (B) 0 only

(C) $\pm 1$ only (D) 1 only

**Q32.** If $A=\begin{pmatrix}1 & -2 & 1\\ 2 & 1 & 3\end{pmatrix}$ and $B=\begin{pmatrix}2 & 1\\ 3 & 2\\ 1 & 1\end{pmatrix}$, then

$(AB)^T$ is equal to -

(A) $\begin{pmatrix}-3 & -2\\ 10 & 7\end{pmatrix}$ (B) $\begin{pmatrix}-3 & 10\\ -2 & 7\end{pmatrix}$

(C) $\begin{pmatrix}-3 & 7\\ 10 & 2\end{pmatrix}$ (D) None of these

**Q33.** If $A=\begin{bmatrix}1 & -2\\ 5 & 3\end{bmatrix}$, then $A+A^T$ equals

(A) $\begin{bmatrix}2 & 3\\ 3 & 6\end{bmatrix}$ (B) $\begin{bmatrix}2 & -4\\ 10 & 6\end{bmatrix}$

(C) $\begin{bmatrix}2 & 4\\ -10 & 6\end{bmatrix}$ (D) None of these

**Q34.** If $A=\begin{bmatrix}0 & 1 & -2\\ -1 & 0 & 5\\ 2 & -5 & 0\end{bmatrix}$, then

(A) $A'=A$ (B) $A'=-A$

(C) $A'=2A$ (D) None of these

**Q35.** An orthogonal matrix is

(A) $\begin{bmatrix}\cos\alpha & 2\sin\alpha\\ -2\sin\alpha & \cos\alpha\end{bmatrix}$

(B) $\begin{bmatrix}\cos\alpha & \sin\alpha\\ -\sin\alpha & \cos\alpha\end{bmatrix}$

(C) $\begin{bmatrix}\cos\alpha & \sin\alpha\\ \sin\alpha & \cos\alpha\end{bmatrix}$

(D) $\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}$

## Symmetric & Skew Symmetric Matrix

**Q36.** The matrix $\begin{bmatrix} 0 & 5 & -7 \\ -5 & 0 & 11 \\ 7 & -11 & 0 \end{bmatrix}$ is known as

(A) Upper triangular matrix
(B) Skew symmetric matrix
(C) Symmetric matrix
(D) Diagonal matrix

**Q37.** If $A$ is a square matrix for which $a_{ij} = i^2 - j^2$, then $A$ is

(A) Zero matrix
(B) Unit matrix
(C) Symmetric matrix
(D) Skew symmetric matrix

**Q38.** If $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$, then $A$ is

(A) Symmetric
(B) Skew-symmetric
(C) Non-singular
(D) Singular

## Adjoint of a Matrix

**Q39.** Adjoint of the matrix $N = \begin{bmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{bmatrix}$ is

(A) $N$ (B) $2N$
(C) $-N$ (D) None of these

**Q40.** If $k$ is a scalar and $I$ is a unit matrix of order 3, then $adj(kI) =$

(A) $k^3 I$ (B) $k^2 I$
(C) $-k^3 I$ (D) $-k^2 I$

**Q41.** If $A$ is a $n \times n$ matrix, then $adj(adj\ A) =$

(A) $|A|^{n-1} A$

(B) $|A|^{n-2} A$

(C) $|A|^n\ n$

(D) None of these

**Q42.** If $A$ is a non- singular matrix, then $A(adj\ A) =$

(A) $A$ (B) $I$
(C) $|A|I$ (D) $|A|^2\ I$

## Inverse of a Matrix

**Q43.** If matrix $A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$, then

(A) $A' = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

(B) $A^{-1} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$

(C) $A.\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = 2I$

(D) $\lambda A = \begin{bmatrix} \lambda & -\lambda \\ 1 & -1 \end{bmatrix}$ where $\lambda$ is a non zero scalar

**Q44.** Let $A = \begin{pmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$, the only correct statement about the matrix $A$ is

(A) $A^2 = I$

(B) $A = (-1)I$, where $I$ is a unit matrix

(C) $A^{-1}$ does not exist

(D) $A$ is a zero matrix

**Q45.** If $A=\begin{bmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{bmatrix}$, then which of the following statements is not correct

(A) $A$ is orthogonal matrix

(B) $A'$ is orthogonal matrix

(C) Determinant $A = 1$

(D) $A$ is not invertible

**Q46.** Inverse of the matrix $\begin{bmatrix}3 & -2 & -1 \\ -4 & 1 & -1 \\ 2 & 0 & 1\end{bmatrix}$ is

(A) $\begin{bmatrix}1 & 2 & 3 \\ 3 & 3 & 7 \\ -2 & -4 & -5\end{bmatrix}$ (B) $\begin{bmatrix}1 & -3 & 5 \\ 7 & 4 & 6 \\ 4 & 2 & 7\end{bmatrix}$

(C) $\begin{bmatrix}1 & 2 & 3 \\ 2 & 5 & 7 \\ -2 & -4 & -5\end{bmatrix}$ (D) $\begin{bmatrix}1 & 2 & -4 \\ 8 & -4 & -5 \\ 3 & 5 & 2\end{bmatrix}$

**Q47.** If $A$ and $B$ are non-singular matrices, then

(A) $(AB)^{-1} = A^{-1}B^{-1}$ (B) $AB = BA$

(C) $(AB)' = A'B'$ (D) $(AB)^{-1} = B^{-1}A^{-1}$

**Q48.** If $A=\begin{bmatrix}i & 0 \\ 0 & i/2\end{bmatrix}$ $(i=\sqrt{-1})$, then $A^{-1}=$

(A) $\begin{bmatrix}i & 0 \\ 0 & i/2\end{bmatrix}$ (B) $\begin{bmatrix}-i & 0 \\ 0 & -2i\end{bmatrix}$

(C) $\begin{bmatrix}i & 0 \\ 0 & 2i\end{bmatrix}$ (D) $\begin{bmatrix}0 & i \\ 2i & 0\end{bmatrix}$

**Q49.** If a matrix $A$ is such that $3A^3+2A^2+5A+I=O$, then its inverse is

(A) $-(3A^2+2A+5I)$ (B) $3A^2+2A+5I$

(C) $3A^2-2A-5I$ (D) None of these

**Q50.** If A, B and C are $n \times n$ matrices and det(A) = 2, det(B) = 3 and det(C) = 5, then the value of the det($A^2BC^{-1}$) is equal to

(A) $\frac{6}{5}$ (B) $\frac{12}{5}$

(C) $\frac{18}{5}$ (D) $\frac{24}{5}$

## Special Matrix

**Q51.** If $A$ is involutory matrix and $I$ is unit matrix of same order, then $(I-A)(I+A)$ is

(A) Zero matrix (B) $A$

(C) $I$ (D) $2A$

**Q52.** The matrix $A=\begin{bmatrix}1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & -1/\sqrt{2}\end{bmatrix}$ is

(A) Involutory (B) Orthogonal

(C) Nilpotent (D) Involutory

**Q53.** The matrix $A=\begin{bmatrix}1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4\end{bmatrix}$ is nilpotent of index

(A) 2 (B) 3

(C) 4 (D) 6

**Q54.** The matrix $A=\frac{1}{3}\begin{bmatrix}1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1\end{bmatrix}$ is

(A) Orthogonal (B) Involutory

(C) Idempotent (D) Nilpotent

## ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

**Q1.** If $D_1$ & $D_2$ are two 3×3 diagonal matrices, then-
(A) $D_1 D_2$ is a diagonal matrix
(B) $D_1D_2 = D_2D_1$
(C) $D_1D_2 + D_2D_1$ is a diagonal matrix
(D) none of these

**Q2.** If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$, then
(A) adj (adj A) = A (B) | adj (adj A) | = 1
(C) | adj A | = 1 (D) None of these

**Q3.** If A and B are two invertible matrices of the same order, then adj (AB) is equal to-
(A) adj B adj A (B) $|B||A|B^{-1}A^{-1}$
(C) $|B||A|A^{-1}B^{-1}$ (D) $|A||B|(AB)^{-1}$

**Q4.** If A and B are symmetric non singular matrices, AB = BA and $A^{-1}B^{-1}$ exist then -
(A) $A^{-1}B$ is symmetric
(B) $AB^{-1}$ is symmetric
(C) $A^{-1}B^{-1}$ is symmetric
(D) None of the above TRUE

## PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

**Paragraph # 1 (Ques. 5 to 7)**

Let A and B are two matrices of same order 3 × 3,

where $A = \begin{pmatrix} 1 & 3 & \lambda+2 \\ 2 & 4 & 8 \\ 3 & 5 & 10 \end{pmatrix}$, $B = \begin{pmatrix} 3 & 1 & 4 \\ 3 & 2 & 5 \\ 2 & 1 & 4 \end{pmatrix}$.

**Q5.** tr(A + B) is equal to
(A) 6 (B) 12
(C) 24 (D) 17

**Q6.** If matrix 2A + 3B is singular, then the value of 2λ is
(A) 6 (B) 12
(C) 24 (D) 17

**Q7.** If λ = 3, then $\frac{1}{7}(\text{tr}(AB) + \text{tr}(BA))$ is equal
(A) 234/7 (B) 233/7
(C) 231/7 (D) 232/7

**Paragraph # 2 (Ques. 8 to 10)**

If A is a symmetric and B skew symmetric matrix and A + B is non singular and $C = (A + B)^{-1}(A - B)$ then:

**Q8.** $C^T(A + B)C =$
(A) A + B (B) A – B
(C) A (D) B

**Q9.** $C^T(A - B)C =$
(A) A + B (B) A – B
(C) A (D) B

**Q10.** $C^TAC$
(A) A + B (B) A – B
(C) A (D) B

**Paragraph # 3 (Ques. 11 to 14)**

If A and B are two matrices of same order 3 × 3,

where $A=\begin{bmatrix}1 & 2 & 3\\ 2 & 3 & 4\\ 5 & 6 & 8\end{bmatrix}$ and $B=\begin{bmatrix}3 & 2 & 5\\ 2 & 3 & 8\\ 7 & 2 & 9\end{bmatrix}$.

**Q11.** The value of Adj (Adj A) is equal to-

(A) –A (B) A

(C) 8A (D) 16A

**Q12.** The value |Adj (Adj A)| is equal to-

(A) 9 (B) 1

(C) –1 (D) 81

**Q13.** The value of |Adj B| is equal to-

(A) 24 (B) $(24)^2$

(C) $(24)^3$ (D) $(8)^2$

**Q14.** The value of |Adj(AB)| is equal to-

(A) 24 (B) $(24)^2$

(C) $(24)^3$ (D) 65

## MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

**Q15.**

| **Column-I** | **Column-II** |
|---|---|
| (A) If $A=\begin{bmatrix}1 & 3\\ 3 & 4\end{bmatrix}$ and $A^2-KA-5I_2=0$ then the value of K is | (P) 4 |
| (B) If $A=\begin{bmatrix}4 & x-2\\ 2x-3 & x+1\end{bmatrix}$ is symmetric, then value of x is | (Q) 2 |
| (C) If A is a square matrix of order 3 and det. A =500, then det. $\left(\frac{A}{5}\right)=$ | (R) 1 |
| | (S) 5 |

**Q16.** Consider a square matrix A of order 2 which has its elements as 0, 1, 2 and 4. Let N denote the number of such matrices, all elements of which are distinct.

| **Column-A** | **Column-B** |
|---|---|
| (A) Possible non-negative value of det(A) is | (P) 2 |
| (B) Sum of values of determinants corresponding to N matrices is | (Q) 4 |
| (C) If absolute value of (det(A)) is least, then possible value of \|adj(adj(adjA)))\| | (R) –2 |
| (D) If det(A) is algebraically least, then possible value of $\det(4A^{-1})$ is | (S) 0 |
| | (T) 8 |

## TRIPLE MATCH COLUMN TYPE

*This section contains ONE table having 3 columns and 4 rows. Based on the table each question has FOUR options [A], [B], [C] AND [D]. **ONLY ONE** of these four is correct.*

# **Column – I contains matrix A of order 3**

# **Column – II contains det (2A)**

# **Column – III contains det (adj A)**

| **Column – I** | **Column – II** | **Column –III** |
|---|---|---|
| (I) $\begin{bmatrix} 1 & 2 & 3 \\ 5 & -1 & -1 \\ 3 & 2 & -1 \end{bmatrix}$ | (P) 48 | (i) 2500 |
| (II) $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 3 & 2 & -1 \end{bmatrix}$ | (Q) 400 | (ii) 225 |
| (III) $\begin{bmatrix} 1 & -2 & -3 \\ -4 & -2 & -3 \\ 3 & 4 & 1 \end{bmatrix}$ | (R) – 120 | (iii) 2116 |
| (IV) $\begin{bmatrix} 28 & 5 & 8 \\ 18 & 5 & 7 \\ 3 & 0 & 0 \end{bmatrix}$ | (S) 368 | (iv) 36 |

**Q17.** Which of following is correct combination:-

(A) II (P) (iii)

(B) I(S) (iii)

(C) III (R) (i)

(D) IV (Q) (ii)

**Q18.** Which of following is correct combination:-

(A) I S (ii)

(B) III(Q) (iii)

(C) II (P) (iv)

(D) IV (R) (i)

**Q19.** Which of following is correct combination:-

(A) I (P) (i)

(B) II(R) (iii)

(C) IV Q (iv)

(D) III Q (i)

## NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

**Q20.** Find the number of $2 \times 2$ matrix satisfying following conditions :

(i) $a_{ij}$ is 1 or –1 ;

(ii) $a_{11}a_{21} + a_{12}a_{22} = 0$

**Q21.** A is a square matrix of order n.

$\ell$ = maximum number of distinct entries if A is a triangular matrix

m = maximum number of distinct entries if A is a diagonal matrix

p = minimum number of zeroes if A is a triangular matrix.

If $\ell + 5 = p + 2m$, find the order of the matrix.

**Q22.** Let $D_k$ is the k × k matrix with 0's in the main diagonal, unity as the element of 1st row and $(f(k))^{th}$ column and k for all other entries. If $f(x) = x - \{x\}$ where $\{x\}$ denotes the fractional part function then the value of det. $(D_2)$ + det. $(D_3)$ equals.

**Q23.** If the product of n matrices

$$\begin{bmatrix}1 & 1\\0 & 1\end{bmatrix}\begin{bmatrix}1 & 2\\0 & 1\end{bmatrix}\begin{bmatrix}1 & 3\\0 & 1\end{bmatrix}.....\begin{bmatrix}1 & n\\0 & 1\end{bmatrix}$$

is equal to the matrix $\begin{bmatrix}1 & 378\\0 & 1\end{bmatrix}$

then the value of n is equal to -

**Q24.** Let $M = \begin{bmatrix}a & -360\\b & c\end{bmatrix}$, where a, b and c are integers. Find the smallest positive value of b such that $M^2 = \mathbf{O}$, where **O** denotes 2 × 2 null matrix.

**Q25.** If A and B are square matrices of order 3, where $|A| = -2$ and $|B| = 1$, then find $\left|(A^{-1})\text{adj}(B^{-1})\text{adj}(2A^{-1})\right|$.

**Q26.** Let $A = \begin{bmatrix}2 & 0 & 7\\0 & 1 & 0\\1 & -2 & 1\end{bmatrix}$

and $B = \begin{bmatrix}-x & 14x & 7x\\0 & 1 & 0\\x & -4x & -2x\end{bmatrix}$

are two matrices such that $AB = (AB)^{-1}$ and $AB \neq I$

(where I is an identity matrix of order 3 × 3).

Find the value of

Tr. $\left(AB + (AB)^2 + (AB)^3 + ..... + (AB)^{100}\right)$,

where Tr. (A) denotes the trace of matrix A.

**Q27.** If A is an idempotent non-zero matrix and I is an identity matrix of the same order, find the value of n, $n \in N$, such that $(A + I)^n = I + 127\,A$.

**Q28.** If A is a diagonal matrix of order 3 such that $A^2 = A$, find number of possible matrices A.

**Q29.** $A = \begin{bmatrix}1 & \tan x\\-\tan x & 1\end{bmatrix}$ then let us define a function $f(x) = \text{det.}\,(A^TA^{-1})$ then find the value of $\underbrace{f(f(f(f.....f(x))))}_{n\text{ times}}$ is $(n \geq 2)$.

## SUBJECTIVE TYPE

**Q30.** If $A = \begin{bmatrix}1 & 2 & 0\\2 & -1 & 0\\0 & 0 & -1\end{bmatrix}$,

show that $5A^{-1} = A^2 + A - 5I$

**Q31.** Prove that if A is non-singular matrix such that A is symmetric then $A^{-1}$ is also symmetric.

**Q32.** Show that $\begin{bmatrix}1 & 1 & 3\\5 & 2 & 6\\-2 & -1 & -3\end{bmatrix}$ is nilpotent matrix of order 3.

## Exercise - 2 — Brain Booster

### SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

**Q1.** If the matrix $\begin{bmatrix} (x-a)^2 & (x-b)^2 & (x-c)^2 \\ (y-a)^2 & (y-b)^2 & (y-c)^2 \\ (z-a)^2 & (z-b)^2 & (z-c)^2 \end{bmatrix}$ is a null matrix, then a, b, c, x, y, z are connected by

(A) $a+b+c=0, x+y+z=0$

(B) $a+b+c=0, x=y=z$

(C) $a=b=c, x+y+z=0$

(D) None of these

**Q2.** The trace $T_r(A)$ of a $3\times3$ matrix $A=(a_{ij})$ is defined by the relation $T_r(A)=a_{11}+a_{22}+a_{33}$ (i.e. $T_r(A)$ is sum of the main diagonal elements). Which of the following statements cannot hold ?

(A) $T_r(kA)=k\,T_r(A)$ (k is a scalar)

(B) $T_r(A+B)=T_r(A)+T_r(B)$

(C) $T_r(I_3)=3$

(D) $T_r(A^2)=(Tr(A))^2$

**Q3.** Let $A=\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$ and $B=\begin{pmatrix}a & 0\\ 0 & b\end{pmatrix}$, a, b ∈ N. Then-

(A) there exist more than one but finite number of B's such that AB = BA

(B) there exist exactly one B such that AB = BA

(C) there exist infinitely many B's such that AB=BA

(D) there cannot exist any B such that AB = BA

**Q4.** The value of x, so that

$$[1\ x\ 1]\begin{bmatrix}1 & 3 & 2\\ 0 & 5 & 1\\ 0 & 3 & 2\end{bmatrix}\begin{bmatrix}1\\ 1\\ x\end{bmatrix}=0 \text{ is}$$

(A) $\frac{-7\pm\sqrt{35}}{2}$

(B) $\frac{-9\pm\sqrt{53}}{2}$

(C) $\pm2$

(D) 0

**Q5.** If A and B are square matrices of order 3 such that $|A|=-1, |B|=3$, then $|3AB|$ equals

(A) –9 (B) –81

(C) –27 (D) 81

**Q6.** Given a third order matrix

$$A=\begin{bmatrix}a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\end{bmatrix}, \text{ If } |A-\lambda I|=0,$$

then the value of

$$\begin{vmatrix}(b_2-\lambda)(c_3-\lambda)-b_3c_2 & b_3c_1-b_1(c_3-\lambda)\\ a_3c_2-a_2(c_3-\lambda) & (a_1-\lambda)(c_3-\lambda)-a_3c_1\\ a_2b_3-a_3(b_2-\lambda) & a_3b_1-b_3(a_1-\lambda)\end{vmatrix}$$

$$\begin{vmatrix}b_1c_2-c_1(b_2-\lambda)\\ a_2c_1-c_2(a_1-\lambda)\\ (a_1-\lambda)(b_2-\lambda)-a_2b_1\end{vmatrix} \text{ is}$$

(A) zero

(B) $a_1(b_2c_3-b_3c_2)-a_2(b_1c_3-b_3c_1)\backslash$

$+a_3(b_1c_2-b_2c_1)$

(C) $\Sigma a_1a_2+\Sigma b_1b_2+\Sigma c_1c_2$

(D) None of these

**Q7.** If $A=\begin{bmatrix} 0 & -\tan\frac{\alpha}{2} \\ \tan\frac{\alpha}{2} & 0 \end{bmatrix}$ and $I_2=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ then $I_2+A=(I_2-A)M$, where M is given by

(A) $\begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & -\cos\alpha \end{bmatrix}$

(B) $\begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$

(C) $\begin{bmatrix} \sin\alpha & -\cos\alpha \\ \cos\alpha & \sin\alpha \end{bmatrix}$

(D) None of these

**Q8.** Matrix A such that $A^2=2A-I$, where I is the identity matrix. Then for $n\geq 2, A^n$ is equal to

(A) $nA-(n-1)I$ (B) $nA-I$

(C) $2^{n-1}A-(n-1)I$ (D) $2^{n-1}A-I$

**Q9.** If $\begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix}$ is orthogonal matrix, then $\alpha^2\beta^2\gamma^2$ equals -

(A) $\frac{1}{2}$ (B) $\frac{1}{6}$

(C) $\frac{1}{36}$ (D) None of these

**Q10.** If A is symmetric matrix and B is a skew symmetric matrix, then for any $n\in N$, which of the followingis not correct?

(A) $A^n$ is symmetric

(B) $A^n$ is symmetric if n is odd only

(C) $B^n$ is skew symmetric if n is odd

(D) $B^n$ is symmetric if n is even

**Q11.** If $P=\begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $A=\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q=PAP^T$ and $x=P^T Q^{2005} P$, then x is equal to -

(A) $\begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} 4+2005\sqrt{3} & 6015 \\ 2005 & 4-2005\sqrt{3} \end{bmatrix}$

(C) $\frac{1}{4}\begin{bmatrix} 2+\sqrt{3} & 1 \\ -1 & 2-\sqrt{3} \end{bmatrix}$

(D) $\frac{1}{4}\begin{bmatrix} 2005 & 2-\sqrt{3} \\ 2+\sqrt{3} & 2005 \end{bmatrix}$

**Q12.** If $A=\begin{bmatrix} 1 & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$ then $|\text{adj (adj A)}|$ is

(A) $(14)^4$ (B) $(14)^3$

(C) $(14)^2$ (D) 14

**Q13.** If $3A=\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ x & 2 & y \end{bmatrix}$ and $AA'=I$, then x + y is equal to -

(A) –5 (B) –4 (C) –3 (D) –2

**Q14.** If $A^2-A+I=0$, then the inverse of A -

(A) $I-A$ (B) $A-I$ (C) A (4) $A+I$

**Q15.** $A=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$, $I=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $A^{-1}=\frac{1}{6}(A^2+cA+dI)$, then the value of c and d are -

(A) –6, –11 (B) 6, 11

(C) –6, 11 (D) 6, –11

**Q16.** Consider the matrix $A=\begin{bmatrix}1 & x & x^2\\ 1 & y & y^2\\ 1 & z & z^2\end{bmatrix}$.

Then A is invertible

(A) For all values of x, y, z

(B) only if atleast two out of x, y, z are equal

(C) only if $x=y=z$

(D) only if x, y and z are distinct

**Q17.** If $A(x_1)=\begin{bmatrix}\cos x_1 & -\sin x_1 & 0\\ \sin x_1 & \cos x_1 & 0\\ 0 & 0 & 1\end{bmatrix}$ and

$A(x_2)=\begin{bmatrix}\cos x_2 & 0 & \sin x_2\\ 0 & 1 & 0\\ -\sin x_2 & 0 & \cos x_2\end{bmatrix}$,

then $(A(x_1).A(x_2))^{-1}$ is equal to

(A) $A(x_1).A(x_2)$ (B) $A(-x_1).A(-x_2)$

(C) $-A(x_2).A(x_1)$ (D) $A(-x_2).A(-x_1)$

**Q18.** If $\begin{bmatrix}a & b\\ c & d\end{bmatrix}\begin{bmatrix}1 & 2\\ 3 & 5\end{bmatrix}=\begin{bmatrix}5 & 1\\ 0 & 2\end{bmatrix}$ then det. $\begin{vmatrix}a & b\\ c & d\end{vmatrix}$ equals

(A) –98 (B) –10

(C) 0 (D) 98

**Q19.** Inverse of the matrix $A=\begin{bmatrix}1 & 3 & 3\\ 1 & 4 & 3\\ 1 & 3 & 4\end{bmatrix}$ equals

(A) Adj A (B) 2 Adj A

(C)A (D) $A^T$

**Q20.** Let $A=\begin{bmatrix}2\sin^2\theta & 0\\ \sin^2\theta & \sin^2\theta\end{bmatrix}$

and $A^{-1}=\begin{bmatrix}1 & 0\\ -1 & 2\end{bmatrix}$

then the most general values of θ are where n is an integer.

(A) $n\pi+(-1)^n\frac{\pi}{4}$ (B) $n\pi\pm\frac{\pi}{3}$

(C) $n\pi\pm\frac{\pi}{4}$ (D) Not possible

**Q21.** The matrix $\begin{bmatrix}1 & 2 & 3\\ 1 & 2 & 3\\ -1 & -2 & -3\end{bmatrix}$ is

(A) idempotent (B) nilpotent

(C) involutory (D) orthogonal

**Q22.** If A is an idempotent matrix satisfying, $(I-0.4A)^{-1}=I-\alpha A$ where I is the unit matrix of the same order as that of A then the value of α is equal to ($\alpha\in R$)

(A) 2/5 (B) 2/3

(C) –2/3 (D) 1/2

**Q23.** If inverse of the matrix $\begin{bmatrix}a_1 & b_1 & c_1\\ a_2 & b_2 & c_2\\ a_3 & b_3 & c_3\end{bmatrix}$ is

$\frac{1}{67}\begin{bmatrix}-6 & 17 & 13\\ 14 & 5 & -8\\ -15 & 9 & -1\end{bmatrix}$

then solution of system of linear equation

$a_1x+b_1y+c_1z=-4;\quad a_2x+b_2y+c_2z=2$

and $a_3x+b_3y+c_3z=11$ is

(A) $x=3, y=-2, z=1$

(B) $x=1, y=2, z=3$

(C) $x=1, y=-2\ z=-3$

(D) cannot determine from given data

## ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

**Q1.** If A and B are square matrices of the same order such that $A^2 = A, B^2 = B, AB = BA = O$, then

(A) $(A+B)^2 = A+B$ (B) $AB^2 = O$

(C) $(A-B)^2 = A-B$ (D) None of these

**Q2.** Let $A = a_{ij}$ be a matrix of order 3 where

$$a_{ij} = \begin{cases} x & \text{if} \quad i = j, x \in R \\ 1 & \text{if} \quad |i-j| = 1 \\ 0 & \text{otherwise} \end{cases}$$, then

which of the following hold(s) good ?

(A) for x = 2, A is a diagonal matrix

(B) A is a symmetric matrix

(C) for x = 2, det A has the value equal to 6

(D) Let f(x) = det A, then the function f (x) has both the maxima and minima

**Q3.** If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$, then

(A) adj (adj A) = A (B) |adj (adj A)| = 1

(C) |adj A| = 1 (D) None of these

**Q4.** If $A^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -1 \end{bmatrix}$, then

(A) |A| = 2

(B) A is non-singular

(C) Adj.A $= \begin{bmatrix} 1/2 & -1/2 & 0 \\ 0 & -1 & 1/2 \\ 0 & 0 & -1/2 \end{bmatrix}$

(D) A is skew symmetric matrix

**Q5.** If $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$, then

(A) Adj A is a zero matrix

(B) Adj $A = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$

(C) $A^{-1} = A$

(D) $A^2 = I$

**Q6.** If A is an invertible matrix then $(\text{adj.A})^{-1}$ is equal to

(A) adj.$(A^{-1})$ (B) $\frac{A}{\det.A}$

(C) A (D) (det. A) A

**Q7.** Let $A_\alpha = \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then

(A) $A_{\alpha+\beta} = A_\alpha A_\beta$ (B) $A_\alpha^{-1} = A_{-\alpha}$

(C) $A_\alpha^{-1} = -A_\alpha$ (D) $A_\alpha^2 = -I$.

**Q8.** Let

$$A = \begin{pmatrix} 0 & \sin\alpha & \sin\alpha\sin\beta \\ -\sin\alpha & 0 & \cos\alpha\cos\beta \\ -\sin\alpha\sin\beta & -\cos\alpha\cos\beta & 0 \end{pmatrix},$$

then

(A) |A| is independent of $\alpha$ and $\beta$

(B) $A^{-1}$ depends only on $\alpha$

(C) $A^{-1}$ depends only on $\beta$

(D) None of these

**Q9.** Suppose $a_1, a_2, \ldots$ real numbers, with $a_1 \neq 0$. If $a_1, a_2, a_3 \ldots$ are in A.P.,then

(A) $A = \begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_5 & a_6 & a_7 \end{bmatrix}$ is singular

(B) The system of equations

$a_1x + a_2y + a_3z = 0$,

$a_4x + a_5y + a_6z = 0$,

$a_7x + a_8y + a_9z = 0$

has infinite number of solutions

(C) $B = \begin{bmatrix} a_1 & ia_2 \\ ia_2 & a_1 \end{bmatrix}$ is non singular

(D) $B = \begin{bmatrix} a_1 & ia_2 \\ ia_2 & a_1 \end{bmatrix}$ is singular

## PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONE OR MORE THAN ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

**Paragraph # 1 (Ques. 10 to 12)**

3 × 3 skew symmetric matrices are formed using 7 elements 1, 1, –1, 0, 0, 0, 0 & remaining 2 elements from set {–1, 0, 1}.

**Q10.** Total number of distinct matrices formed

(A) 6 (B) 8

(C) 12 (D) 10

**Q11.** Number of matrices that can be formed in which $a_{12} = a_{32}$, are

(A) 2 (B) 6

(C) 4 (D) 0

**Q12.** The number of skew-symmetric matrices in which $a_{12} = a_{31}$ is K. Now, in these 'K' matrices, if element $a_{33}$ is replaced by 2, then sum of determinant values of these K matrices is

(A) 2 (B) 4

(C) 0 (D) –2

**Paragraph # 2 (Ques. 13 to 14)**

A Pythagorean triplet of positive integers [a,b,c] such that $a^2 + b^2 = c^2$.

Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 2 & 2 \\ -2 & -1 & -2 \\ 2 & 2 & 3 \end{bmatrix}$ $C = \begin{bmatrix} -1 & -2 & -2 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{bmatrix}$

**Q13.** Which is/are Pythagorean triplets?

(A) [3 4 5] A (B) [3 4 5]B

(C) [3 4 5]C (D) None of these

**Q14.** Which of the following holds good?

(A) $A^{-1} = \text{adj } A$

(B) $(AB)^{-1} = \text{adj } (A - B)$

(C) $(BC)^{-1} = \text{adj}(BC)$

(D) $(ABC)^{-1} \neq \text{adj}(ABC)$

**Paragraph # 3 (Ques. 15 to 17)**

$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$, if $U_1$, $U_2$ and $U_3$ are columns matrices satisfying. $AU_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $AU_2 = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$,

$AU_3 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$ and U is 3 × 3 matrix whose columns are $U_1$, $U_2$, $U_3$ then answer the following questions -

**Q15.** The value of |U| is -

(A) 3 (B) –3 (C) 3/2 (D) 2

**Q16.** The sum of the elements of $U^{-1}$ is -

(A) –1 (B) 0

(C) 1 (D) 3

**Q17.** The value of $\begin{bmatrix} 3 & 2 & 0 \end{bmatrix} U \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$ is -

(A) [5] (B) [5/2]

(C) [4] (D) [3/2]

**Paragraph # 4 (Ques. 18 to 20)**

Let P(x, y) be any point and P'($x_1$, $y_1$) be its image in x-axis then

$x_1 = x$

$y_1 = -y$

This system of equation is equivalent to the matrix equation.

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = A \begin{bmatrix} x \\ y \end{bmatrix}$$

where A is a square matrix of order 2

Similarly $\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = B \begin{bmatrix} x \\ y \end{bmatrix}$, $\begin{bmatrix} x_3 \\ y_3 \end{bmatrix} = C \begin{bmatrix} x \\ y \end{bmatrix}$, $\begin{bmatrix} x_4 \\ y_4 \end{bmatrix} = D \begin{bmatrix} x \\ y \end{bmatrix}$

represents the reflection of point (x, y) in y-axis, origin and the line y = x respectively.

**Q18.** The value of A + B + C + D is -

(A) $\begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix}$ (B) $\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ (D) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

**Q19.** Let X be a square matrix given by

$X = A + AD^2 + AD^4 + \ldots\ldots\ldots + AD^{2n-2}$,

then X is -

(A) $\begin{bmatrix} -n & 0 \\ 0 & -n \end{bmatrix}$ (B) $\begin{bmatrix} n & 0 \\ 0 & n \end{bmatrix}$

(C) $\begin{bmatrix} n & 0 \\ 0 & -n \end{bmatrix}$ (D) $\begin{bmatrix} -n & 0 \\ 0 & n \end{bmatrix}$

**Q20.** Let P(a, b) be a point & $\begin{bmatrix} x \\ y \end{bmatrix} = DCBA \begin{bmatrix} a \\ b \end{bmatrix}$

then Q(x, y) represents the reflection of point P(a, b) in-

(A) x-axis (B) y-axis

(C) origin (D) line y = x

## MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

**Q21.**

| **Column-I** | **Column-II** |
|---|---|
| (A) If A is a square matrix of order 3 and det A = 162 then $\det\left(\frac{A}{3}\right)$ = | (P) 6 |
| (B) If A is a matrix such that $A^2 = A$ and $(I + A)^5 = I + \lambda A$ then $\frac{2\lambda + 1}{7}$ | (Q) 5 |
| (C) If $A = \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}$ and $A^2 - xA + yI = 0$ then y − x = | (R) 0 |
| (D) If $A = \begin{bmatrix} 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \\ 17 & 18 & 19 & 20 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 3 & 5 & 7 \\ -3 & -3 & -10 & -10 \\ 5 & 10 & 5 & 0 \\ 7 & 10 & 0 & 7 \end{bmatrix}$ then $(AB)_{23}$ | (S) 9 |

**Q22.**

| Column-I | Column-II |
|---|---|
| (A) If a, b and c are all different from zero such that $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=0$, then the $A=\begin{pmatrix}1+a & 1 & 1\\ 1 & 1+b & 1\\ 1 & 1 & 1+c\end{pmatrix}$ is | (P) symmetric matrix |
| (B) If $\alpha, \beta, \gamma$ are three real numbers, then the matrix $A=\begin{pmatrix}1 & \cos(\alpha-\beta) & \cos(\alpha-\gamma)\\ \cos(\beta-\alpha) & 1 & \cos(\beta-\gamma)\\ \cos(\gamma-\alpha) & \cos(\gamma-\beta) & 1\end{pmatrix}$ is | (Q) singular |
| (C) If A, B and C are the angles of a triangle, then the matrix $A=\begin{pmatrix}\sin 2A & \sin C & \sin B\\ \sin C & \sin 2B & \sin A\\ \sin B & \sin A & \sin 2C\end{pmatrix}$ is | (R) non singular |
| | (S) invertible |
| | (T) non invertible |

## NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

**Q23.** Let $A=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$ and $B=\begin{bmatrix}p\\ q\end{bmatrix}\neq\begin{bmatrix}0\\ 0\end{bmatrix}$.

Such that AB = B and a + d = 5050.

Find the value of (ad – bc).

**Q24.** If the matrices $A=\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$ and $B=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$

(a, b, c, d not all simultaneously zero)

commute, find the value of $\frac{d-b}{a+c-b}$.

**Q25.** Let $A=\begin{bmatrix}3x^2\\ 1\\ 6x\end{bmatrix}$, $B=[a\ b\ c]$

and $C=\begin{bmatrix}(x+2)^2 & 5x^2 & 2x\\ 5x^2 & 2x & (x+2)^2\\ 2x & (x+2)^2 & 5x^2\end{bmatrix}$

be three given matrices, where a, b, c and x $\in$ R. Given that tr(AB) = tr(C) $\forall$ x $\in$ R, where tr(A) denotes trace of A.

Find the value of (a + b + c)

**Q26.** Let $A=\begin{bmatrix}1 & \frac{3}{2}\\ 1 & 2\end{bmatrix}$, $B=\begin{bmatrix}4 & -3\\ -2 & 2\end{bmatrix}$

and $C_r=\begin{bmatrix}r.3^r & 2^r\\ 0 & (r-1)3^r\end{bmatrix}$

be 3 given matrices. Compute the value of $\sum_{r=1}^{50}\text{tr.}\left((AB)^rC_r\right)$. (where tr.(A) denotes trace of matrix A)

**Q27.** If $A=\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}$ and $\det(A^n - I) = 1 - \lambda^n$, $n \in N$, then $\lambda$ is equal to

**Q28.** Let A be the set of all $3 \times 3$ skew symmetric matrices whose entries are either $-1$, 0 or 1. If there are exactly three 0's, three 1's and three $(-1)$'s, then find the number of such matrices.

**Q29.** $A_{3\times 3}$ is a matrix such that $|A|=a$, $B = (\text{adj } A)$ such that $|B|= b$.
Find the value of $(ab^2 + a^2b + 1)S$ where

$$\frac{1}{2}S=\frac{a}{b}+\frac{a^2}{b^3}+\frac{a^3}{b^5}+\ldots\ldots \text{ up to } \infty, \text{ and } a=3.$$

**Q30.** If A is a square matrix of order 3 such that $|A| = 2$ then find the value of $|(\text{adj } A^{-1})^{-1}|$.

## SUBJECTIVE TYPE

**Q31.** Let $A=\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & 2\\ 2 & 2 & 1\end{bmatrix}$, then show that

$$A^{-1}=\frac{1}{5}(A-4I_3)$$

**Q32.** If $\begin{bmatrix}1 & 2 & a\\ 0 & 1 & 4\\ 0 & 0 & 1\end{bmatrix}^n=\begin{bmatrix}1 & 18 & 2007\\ 0 & 1 & 36\\ 0 & 0 & 1\end{bmatrix}$
then find $a + n$.

**Q33.** If $f(x) = x^2 - 3x + 3$ and $A=\begin{bmatrix}2 & 1\\ -1 & 1\end{bmatrix}$ be a square matrix then prove that $f(A) = \mathbf{O}$. Hence find $A^4$.

**Q34.** If $A=\begin{bmatrix}1 & 0 & 2\\ 0 & 2 & 1\\ 2 & 0 & 3\end{bmatrix}$, then show that the matrix A is a root of the polynomial,
$f(x) = x^3 - 6x^2 + 7x + 2$.

**Q35.** Find the product of two matrices A & B, where

$$A=\begin{bmatrix}-5 & 1 & 3\\ 7 & 1 & -5\\ 1 & -1 & 1\end{bmatrix} \text{ \& } B=\begin{bmatrix}1 & 1 & 2\\ 3 & 2 & 1\\ 2 & 1 & 3\end{bmatrix}$$

and use it to solve the following system of linear equations, $x + y + 2z = 1$; $3x + 2y + z = 7$; $2x + y + 3z = 2$.

**Q36.** Determine the values of a and b for which the system $\begin{bmatrix}3 & -2 & 1\\ 5 & -8 & 9\\ 2 & 1 & a\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}b\\ 3\\ -1\end{bmatrix}$

(A) has a unique solution
(B) has no solution and
(C) has infinitely many solutions

## Answer Key

**Ex-1**

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| C | B | A | D | B | B | B | B | C | A |
| **11** | **12** | **13** | **14** | **15** | **16** | **17** | **18** | **19** | **20** |
| D | C | C | A | C | A | B | D | C | D |
| **21** | **22** | **23** | **24** | **25** | **26** | **27** | **28** | **29** | **30** |
| B | C | A | A | B | B | D | A | D | A |
| **31** | **32** | **33** | **34** | **35** | **36** | **37** | **38** | **39** | **40** |
| C | B | A | B | B | B | D | C | A | B |
| **41** | **42** | **43** | **44** | **45** | **46** | **47** | **48** | **49** | **50** |
| B | C | C | A | D | C | D | B | A | B |
| **51** | **52** | **53** | **54** | | | | | | |
| A | C | A | A | | | | | | |

## Ex - 1A

**1.** A,B,C **2.** A,B,C **3.** A,B,D **4.** A,B,C **5.** C

**6.** D **7.** A **8.** A **9.** B **10.** C

**11.** A **12.** B **13.** B **14.** B

**15.** (A) → (S) ; (B) → (R) ; (C) → (P)

**16.** (A) → (P,Q,T) ; (B) → (S) ; (C) → (P,R) ; (D) → (R)

**17.** B **18.** C **19.** D **20.** 8 **21.** 4

**22.** 34 **23.** 27 **24.** 10 **26.** – 8 **26.** 100

**27.** n = 7 **28.** 7 **29.** 1

**Ex - 2**

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| D | D | C | B | B | A | B | A | C | B |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| A | A | C | A | C | D | D | B | A | C |
| 21 | 22 | 23 | | | | | | | |
| B | C | A | | | | | | | |

## EX - 2A

**1.** A,B **2.** B,D **3.** A,B,C **4.** B,C **5.** B,C,D

**6.** A,B,C **7.** A,B **8.** A **9.** A,B,C **10.** C

**11.** A **12.** B **13.** A,B,C **14.** A,B,C **15.** A

**16.** B **17.** A **18.** B **19.** C **20.** D

**21.** (A) → (P) ; (B) → (S) ; (C) → (Q) ; (D) → (R)

**22.** (A) → (P,R,S); (B) → (P,Q,T); (C) → (P,Q,T);

**23.** 5049 **24.** 1 **25.** 7 **26.** $3(49.3^{50} + 1)$

**27.** 2 **28.** 4 **29.** 3 **30.** 200

**31.** x = 2, y = 1, z = –1

**32.** **(A)** $a \neq -3, b \in R$; **(B)** $a = -3$ and $b \neq 1/3$; **(C)** $a = -3, b = 1/3$

**33.** **(A)** $X = \begin{bmatrix} -3 & -3 \\ \frac{5}{2} & 2 \end{bmatrix}$, **(B)** $X = \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$ **(C)** no solution

**34.** **(A)** $X = \begin{bmatrix} a & b \\ 2-2a & 1-2b \end{bmatrix}$ for a, b ∈ R ; **(B)** X does not exist ;

**(C)** $X = \begin{bmatrix} a & -3a \\ c & -3c \end{bmatrix}$ a, c ∈ R and 3a + c ≠ 0

**35.** $a = -3, \ b = \frac{1}{3}$ **36.** x=2, y=1, z=0

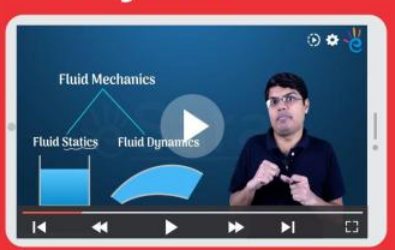
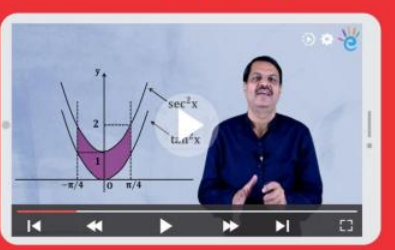
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